

Map Operations Specialist

Navigation, Routing & Just-in-time Alerts

Map Data Quality · OpenStreetMap · Routing Engine · Just-in-time Navigation Alerts



Tasco Building, Hanoi – Pham Hung



TASCO Mapping



Individual Contributor



Reports to Head of Mapping



Min. 3 years experience

ABOUT THE COMPANY

About Tasco

VISION

A technology-enabled investment ecosystem, integrating Smart Mobility, Insurance, and Investments in essential services and innovation.

STRATEGIC PARTNERS

Geely Auto

Mitsui & Co.

IFC – World Bank Group

Corporate Information

Stock Ticker	HUT (HNX)
Established	1971
Subsidiaries & Affiliates	88
Employees	~10,000
Consolidated Sales	\$1.4 billion
Total Assets	\$1.98 billion
Website	tasco.com.vn

BUSINESS ECOSYSTEM

MOBILITY

Tasco Auto
VETC
Carpla
VETC Go

INSURANCE

Tasco Insurance
Tasco Finance

INVESTMENT

DNP Water · Tasco Mall
Ana Mandara Villas
Six Senses
Inochi

WHY TASCO

Our Mobility Ecosystem in Numbers

TASCO AUTO

40k+

New Cars Sold

180

Showrooms

Top 5

Tax-paying Retailers

16

Car Brands

14.8%

Market Share

VETC — ETC & MOBILITY

4.2m

Car Owners

~300

Parking Lots

2m

Transactions / Day

650k

Car Repairs

75%

ETC Market Share

0 Job Overview

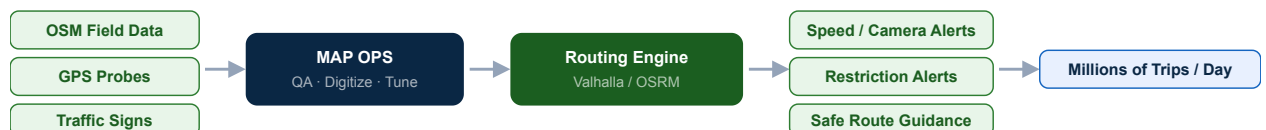
POSITION	Map Operations Specialist
FOCUS	Map Data Quality, Routing & Just-in-time Alerts
DEPARTMENT	TASCO.COM.VN Mapping
LOCATION	Tasco Building Hanoi, Pham Hung
LEVEL	Individual Contributor
REPORTS TO	Head of Mapping

1 Job Purpose

The Map Operations Specialist owns the quality of the digital map system that powers our navigation ecosystem. This role sets data-quality standards for OpenStreetMap (OSM), leads digitization of Vietnam's traffic-sign network, and tunes the routing engine to deliver accurate just-in-time alerts for millions of trips per day.

This is a **high-impact role** that works across Engineering, Data Science, and Product to ensure the navigation experience meets international standards.

HOW THE ROLE WORKS — DATA TO NAVIGATION QUALITY



2 Key Responsibilities

A. Map Data Quality Governance (OpenStreetMap)

- Own and define data-quality standards for OpenStreetMap (OSM) data across all operating regions.
- Design and maintain large-scale data monitoring, cleaning, and QA pipelines.
- Resolve issues in road-network topology, node connectivity, and missing/incorrect tags to keep data routing-ready for Valhalla, GraphHopper, and OSRM.
- Define and track data-quality KPIs by region and road class.

B. Traffic Signs & Restrictions Management

- Lead digitization and continuous update of Vietnam's traffic-sign network: speed limits, enforcement cameras, turn/parking/one-way restrictions, lane info, and warning signs.
- Manage complex traffic rules: turn restrictions, time-based restrictions, and road closures (segment / route / zone).
- Ensure sign data is accurately map-matched into the routing graph to prevent routing into prohibited roads.
- Establish data-collection workflows (cameras, dashcams, GPS probes, user reports) and a continuous update pipeline.

C. Just-in-time Alerts & Navigation Experience

- Build logic and manage spatial data so the system emits accurate real-time alerts (speed limits, cameras, route changes) as vehicles move.
- Partner with Engineering to optimize map-matching (HMM, snap-to-road), minimizing latency and false positives / false negatives.
- Define SLOs/SLAs for alert accuracy and latency; set up monitoring dashboards.

D. Troubleshooting, RCA & Routing Tuning

- Lead resolution of critical positioning/navigation incidents from systems and users (GPS drift, on-ground sign mismatches).
- Conduct root-cause analysis (RCA) for trips with wrong ETA or suboptimal routes.
- Tune routing-engine costing models, penalties, and restrictions based on real-world data.
- Translate insights into actionable improvements for product and data infrastructure.

E. Technical Leadership & Mentorship

- Mentor map operators and junior engineers; review work and raise the team's quality bar.
- Build documentation, playbooks, and standard operating procedures (SOPs) for map-data work.
- Partner with Product, Engineering, and Data Science to shape the map-quality roadmap.

3

Requirements

MUST-HAVE

- ✓ Minimum 3 years of hands-on experience as a Map Operator, GIS Data Engineer, or equivalent at a maps, navigation, ride-hailing, or logistics company.
- ✓ Bachelor's degree in Geomatics, GIS, Computer Science, or a related field.
- ✓ Deep knowledge of Vietnam's road-traffic sign system and OpenStreetMap

NICE-TO-HAVE

- ✓ Experience with map-matching algorithms (HMM, snap-to-road) and large-scale GPS-probe data processing.
- ✓ Familiarity with osm2pgsql, Osmium, or planet-scale OSM tooling.
- ✓ Experience with streaming/real-time systems (Kafka, Flink) for location data.
- ✓ Contributions to the OpenStreetMap community or related open-source

data structures (tags, relations, restrictions).

- ✓ Proficient in QGIS, PostgreSQL/PostGIS, and spatial data analysis tooling.
- ✓ Solid understanding of routing algorithms and at least one open-source routing engine (Valhalla, GraphHopper, or OSRM).
- ✓ Proficient in Python for ETL, automation, and internal tooling.
- ✓ Experience with map editors (JOSM, iD, or RapID) and the Overpass API.
- ✓ Strong English reading comprehension for technical documentation.

projects.

- ✓ Background in Automotive / In-car Navigation / ADAS.

SOFT SKILLS & MINDSET

- ✓ Data-driven mindset; decisions grounded in metrics.
- ✓ Able to balance data quality, delivery speed, and operating cost.
- ✓ Strong communication with both technical and non-technical stakeholders.
- ✓ Takes ownership proactively; works independently and can lead small working groups when needed.

4 Success Metrics — 6 to 12 Months

- ◆ Measurable improvement in OSM data-quality KPIs in priority regions (targets agreed with manager).

- ◆ Reduce false-positive / false-negative rates of just-in-time alerts to target levels.

- ◆ Reduce incidents of routing into restricted or wrong roads.

- ◆ Deliver a scalable sign-digitization pipeline ready for new-city expansion.

- ◆ Build and roll out SOPs across the team.

5 What We Offer

Compensation

Competitive salary, performance bonus, and 13th-month salary.

Health & Insurance

Premium health insurance for self and dependents; statutory insurance per Vietnamese labor law.

Learning Budget

Budget for courses, international conferences, and professional certifications.

Flexible Work

Flexible / hybrid working policy.

Impact

Direct impact on a large-scale Smart Mobility / Automotive product used by millions.

Engineering Culture

Strong engineering culture working on core map and navigation challenges.

How to Apply

Send your email with subject **[Tasco Mapping Application]** with your CV (English or Vietnamese) and a portfolio or descriptions of map / navigation projects you have worked on.

phung.dm@tasco.com.vn

tuyendung@vtii.vn

[Tasco Mapping Application]

OSM contribution links or personal repositories are encouraged.